

System Design Proposal

For

***Digital Park Guide Training Platform for Sarawak's
National Parks and Nature Reserves***

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1.0. Project Background / Problem Description

To provide high-quality experiences for visitors to Sarawak's national parks and nature reserves, park guides are essential in delivering relevant information and promoting tourism. From the past until now, the training programs have been conducted in a traditional way, where information is physically delivered to all park guides. This causes long teaching periods, and older park guides may not receive updated information, which can decrease their performance due to outdated guidance and knowledge. Hence, Sarawak Forestry Corporation will be building a digital training platform that can be accessed through a mobile app and website, providing a more convenient training system for all park guides. Some of the issues and challenges are listed below and are intended to be resolved by using the new digital training platform.

Issues

- Park guides are not accessible to most updated information and knowledge.
- New participants who want to join will be taking time to join all the physical classes.
- Park guides are hard to track their progress and access to their certificate when necessary.

Challenges

- Outdated information will reduce the quality of guidance provided to visitors.
- Traditional training methods will lower the training quality.
- Lack of opportunity for practical training.

To resolve the issues and challenges mentioned above, a digital training platform will be developed through both a mobile application and a website. What the platform does is provide flexibility and updated materials, which allow all park guides to track their enrolment and certification. This platform increases the effectiveness of training for both existing and new park guides.

2.0. Scope

A project with high quality of deliverables is highly depends on the project scope. Project scope outlines the general aims of the project, specific objectives which are the features including in the system itself, certain limitations of the project that can be not included in the system and potential further improvements (out of scope features) that can be considered in the future time.

The main issue that the project aims to solve is to solve the pain points of training park guides due to several reasons such as inflexible training platform. In order to solve this issue, an online training platform that can be worked on both web-based and mobile platform is considered to develop. Through the online training platform, park guides could easily access to the relevant training materials and track their own certification.

2.1. Goals/Aims

According to the problem domain of this project: park guides have limited access to the inflexible physical training platform, and they may have insufficient access to keep themselves equipped with the most updated related information, the goals or aims have been clearly defined as below to facilitate the project's development process.

Goals/Aims: The system designed is compatible in both mobile and web-based applications, aims to help the Sarawak Forestry Corporation (SFC) for creating a digital training platform for the park guides, ensure best practice during the guided tour and eventually giving the visitors the best possible experience in the guided tour. The training platform includes the dashboard to be browsed by the users easily for tracking their learning materials and learning progress. There are two roles in the system: admin and users (trainee and park guides) to enable access control in the system. Additionally, certain cybersecurity measures are implemented to ensure the guided tours are compliant with the safety regulations. AI models are expected to be implemented in order to be used in the body-worn cameras for object-detection.

2.2. Objectives

Scope 1: Develop a Digital Training Platform for Park Guides

- Develop an application that is capable to be used in both mobile and web-based platform. The application can provide interactive/informative learning modules for the park guide training, covering the topics of conservation, biodiversity, eco-tourism, legislation, and safety guidelines.
- Other than learning flexibly, park guides are allowed to track their learning progress, managing their certification, and notifications from admin.

Scope 2: Emphasize cybersecurity measures and data protection

- Implement secure login for both admin and users (trainers and park guides), data encryption and access control of admin and user roles. These features implemented to protect the guide personal information such as profile, training progress and assessment data.
- Cybersecurity measures implemented ensure reliability of the users towards the platform.

Scope 3: Implementation of AI model for enforcing guided tours monitor

- Body-worn cameras to be used the park guide in every guided tour, implemented with AI-models which able to detect abnormal activities to ensure the guided tours are compliant with the safety regulations.
- Alerts will be automatically triggered if abnormal activity is detected. Video streams are automatically saved to be used as the purpose of reporting and reviewing if necessary.

2.3. Constraints and out of scope limitations

Constraints and limitations should be clearly defined in a project to ensure the development process go through well without any distractions. A system designed should keep updating to resolve the issues spotted. At here, limitations could make the system to be more precise to specific goals. Limitations include constraints and out of scope contexts. Constraints refer to the technical restriction during the system development while out of scope contexts refer to the potential further improvement of the system and these are yet to be developed currently.

Constraints:

- Technical constraints: limited internet connectivity in remote park, might cause inconvenient in some features such as the body-worn camera
- AI model to be implemented has the concern of insufficient accuracy if the quality of training data is not comprehensive enough.

Out of Scope features:

- Advanced AI implementation: facial recognition of visitors
- Complex certification authority system: government-recognized certification system
- Payment methods to enable the park guides to pay for accessing the training modules

3.0. Stakeholders

Stakeholders are individuals or groups who have an interest in the outcomes of a business or project. To ensure a clear understanding of the people involved in the project, either directly or indirectly, the table below lists the stakeholders and their interests.

Stakeholders	Interests
Product Owner	The person who communicates with the client to convert the requirement of the client into features that need to be implemented in the system.
Project Managers	The person responsible to organise the project's workload to ensure that all tasks are finished on time as scheduled.
Developers	The person responsible for developing the features required for the project on schedule and within the allocated budget.
Customers/Users	The person who will use and interact with the system directly.
Client	The person who having the issue that need to be solved by the system.
Project Sponsor	The person who will supports the developers by providing the resources, tools or technologies needed for the project.

Table 1. Stakeholders

Some of the stakeholders that have been listed above will interact directly with the system as users, such as the customers and the client will be represented as admin in this case. By aligning the stakeholders' interests with the objectives of this project, some of the features will be accessible only to specific stakeholders. The table below lists the features that will be used by the stakeholders who interact with the system.

Stakeholders	Features
Customers/Users	• Access training modules • Track training progress and performance • View certification • Login/Logout securely • Receive notifications from admin • Receive alert from detection system
Client/Admin	• Manage user accounts • Create and upload training modules • Approve or issue certification • Track training progress • Send announcements or notifications to users • Receive alert from detection system

Table 2. Stakeholders and Their Interactions with The System

4.0. Deliverables and Schedules

This is rather self-explanatory. What is to be delivered? (In a software project, for example, this might include the actual binary, all source code, user manual, training program, etc.) Don't repeat what is in the previous section.

4.1. Deliverables of the Project

Deliverables are any tangible or intangible output produced as a result of project work that can be verified and measured (Victoria Landsmann, 2025). The main deliverable for this project is a functional mobile app and web-based training platform that provide interactive modules, progress tracking, certification management and secure login for the park guides. The system integrated with AI-powered abnormal activity detection to support conservation enforcement. The main deliverable of the project can be broken down into several components listed below:

Hardware

1. Webcam

Software

1. User Dashboard
2. Admin Dashboard
3. AI-based Abnormal Activities Detection System

The hardware listed will be used to simulate the environment of the AI-based abnormal activity system as it provides the same core input as a body-worn camera that is continuous video footage of human activity. Whereas the software listed will be the interface for the users of the project to use.

Other deliverables that will be completed by the end of this project includes the Sprint Report and Final Report that will show the concept of the system and its development process such as the management plan of the project. The deliverables will also include presentation slides of the project that will be used to present and explain the system to the stakeholders.

4.2. Initial Release Schedule

4.2.1. Product Backlogs

To ensure the main deliverables for this project can be completed and functional by the end of the project, product backlogs have been used to list down the features and functionalities that will be implemented in this system. Each item in the product backlogs is crucial to solve the issues and challenges stated above along with achieving the aims of the project.

Product backlogs will provide the details and functionality about the features that will be implemented in the system. The dependencies of each item will show that whether the item will depend on the other items to finish before it can continue to be developed. Other than that, the business value indicates that how valuable the feature is for the stakeholders and guide backlog prioritization. Lastly, the release schedule shows the sprint allocation for each feature to be developed throughout the development phase of the project. The first sprint of the project mainly focuses on building the core foundation of the platform such as the user and admin dashboards. While the second sprint focuses on advanced monitoring and system reliability features. The table below shows the information of the product backlogs for this project.

No.	Item	Functionality	Dependencies	Business Value (1 least – 10 most)	Release Schedule (Sprint #1 2)
1	User and admin dashboard	To provide user and admin an interactable user interface that allows user to access their training modules, keeps track of their training progress and allows them to check their qualifications and certifications plus allow admin to perform user control.	-	10	Sprint #1
2	Training modules	To provide park guides appropriate and standard training materials with quizzes. Admin will be able to check and update training modules and progress.	-	10	Sprint #1
3	User and Admin account system	To control what contents different user level can access, to save users' personal information. Only admin will be able to create account for park guides to avoid unrestricted access.	-	10	Sprint #1
4	User data secure and encryption	To secure user data using encryption measures. With end-to-end encryption, other individuals that are not users or admin will not be able to access the database unless they have their account password since the key is stored in their account.	1, 2, 3	9	Sprint #1
5	AI model training	AI model that is able to recognize abnormal and prohibited activities or situations in a forest.	-	10	Sprint #1
6	AI-based anomaly detection system	Integration of AI model into a system that will save the video recording of incident occurred.	5	9	Sprint #1
7	User notification system	User will be able to receive notification from admin about training module progress and	1, 2, 3	7	Sprint #2

		updates, and admin will receive notification of users' training progress.			
8	Automated alert system	AI will send a signal to the detection system and the system will alert park guides and admin about abnormal activities occurring after AI detected it. The detection system will mark the timeline when anomaly occurs to allow easier and faster access.	6	7	Sprint #2
9	Database backup/restore	The database will automatically backup every 2 days and save up to 5 backups to avoid data loss.	-	8	Sprint #2

Table 3. Initial Release Schedule

Project Timeline

A project involves several phases during the process of development to ensure different aspects are developed and delivered correctly by the end of the project. Establishing a clear project timeline is essential to help developers monitor progress and manage time effectively so that all required modules are implemented successfully. Various project management frameworks can be used to define the project timeline. For this project, the team decided to use the SCRUM methodology with a sprint-based approach to enable a more agile development process.

The whole process of the project will divide into 2 main sprints where Sprint 1 takes 46 days and Sprint 2 takes 20 days. The whole development process will take approximately 88 days or 12 weeks starting from the creation of project proposal.

There will be a sprint planning session before each sprint start. This allows the developers to review back all the things that the team need to know and work on as well as clarify the due date of each task. Items in the product backlogs will be broken down into more details of actions in the sprint planning session. After each sprint ended, the leader will conduct a team meeting to review on the progress of the work and things that had been done. The team will also examine on the leftover work and whether there are any aspects that need to be improved. After the reflection, the team will document the meeting outcome for future reference. The team is expected to meet the stakeholder and project sponsor to show the progress of work and also receive constructive feedback from them.

The prototype is expected to function well at the end of Sprint 2 so that the team can prepare for final documentation of the project. At the end of the project, all items in the product backlog should have been implemented as the main deliverables.

Jira is used to create a Gantt Chart with every item in the product backlog listed together with the duration to complete each of them for better visualization of the project timeline. The team can refer to the Gantt Chart to make sure that all developers are on track. Based on the Gantt Chart, the first sprint will start on 2 March 2026 until 17 April 2026, whereas the second sprint will start on 18 April 2026 and end on 8 May 2026. The remaining time will be reserve for the

project final documentation. The whole project will start from 2 March 2026 until 29 May 2026. The timeline is subjective to change according to the needs and situations of the project thus further notification will be given to the other stakeholders if the timeline shifted for some adjustment.

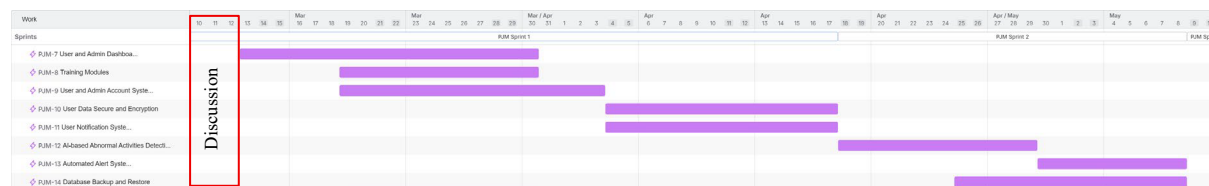


Figure 1. Jira Gantt Chart

5.0. Solution Direction

The chosen solution is a hybrid application that provides a consistent user experience across both web and mobile interfaces. This approach ensures a unified management dashboard for administrators while providing park guides with a mobile tool that supports flexible training resources. To overcome the challenges of limited network connectivity in deep jungle environments, the system utilises Edge-AI processing. Edge-AI means running AI models on devices without relying on cloud connectivity. The body-worn cameras are using local data storage and built-in AI models that detect abnormal activities in real-time. Park guides will be alerted immediately through camera's vibration while park authorities will receive alerts when the device reconnects to a network.

The KoST method splits analysis into three parts: Knowledge, Skill and Technology. From a knowledge perspective, it includes the problem domain and the solution domain. In the problem domain, we should know about park operations and legislation, the eco-tourism industry and wildlife conservation. On the solution domain side, we must be equipped with knowledge about AI models and IoT, cybersecurity, database and digital learning technologies. Besides, we have experience in training AI models, developing web and designing database which support the solution requirements. From a technology perspective, there are some existing cross-platform frameworks such as React Native allow developers to build hybrid applications. Although our project focuses more on natural environments, wearable devices like smartwatches already use Edge AI for human activity recognition in healthcare, showing that this technology is adaptable. Many IoT devices use local storage combined with delayed synchronisation when network connectivity is restored such as drones. It means that there is no technological gap in handling zero-connectivity environments. AI models for detecting abnormal behaviour are already established so this also shows no gap.

Since there are many components in our project such as frontend, backend and AI. Therefore, our analysis of solution direction will split into three parts:

Frontend Framework

Potential Solution	Project Forces	SWOT	KoST	Final Decision
React Native (with Expo)	Time: Plausible Cost: No cost required	Strength: Cross-platform support, fast development and	Knowledge Gap: Moderate as the development requires more	Accepted

	Scope: On scope	often used in project resources Weakness: Performance slightly below native Opportunities: Reuse code for web + mobile Threats: Dependency on external libraries	advanced understanding Skill Gap: Moderate as all team members have experience to work with development of application but in other frameworks Technology Gap: No gap as tools and frameworks required for offline mobile app development are widely available	
Native Mobile Development (iOS/Android separately)	Time: Not plausible Cost: Minimal Scope: On scope	Strength: Best performance & device integration Weakness: Requires two codebases and causes slow development Opportunities: Fine-grained control over offline features Threats: Difficult to maintain and possible to lack consistency in functionalities	Knowledge Gap: High as the development requires more advanced understanding of different platforms Skill Gap: High as all team members must work with the development of application in different and unfamiliar frameworks Technology Gap: Moderate gap as tools and frameworks required for separate mobile app developments are not as widely available	Rejected
Architectures: MVVM/MVC	Time: Plausible Cost: Minimal	Strength: Scalable code, maintainable UI logic Weakness:	Knowledge Gap: Moderate as the team is familiar with modular front-end patterns	Rejected

	Scope: On scope	Extra structure requires discipline Opportunities: Provides more options for developers Threats: Learning curves can be tough for developers	Skill Gap: Moderate as all team members worked with the development of web and software Technology Gap: Low gap as tools and frameworks required for separate mobile app developments are widely available	
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Table 4. Solutions analysis for Mobile Application Component

Backend Framework

Potential Solution	Project Forces	SWOT	KoST	Final Decision
Hybrid Approaches	Time: Plausible Cost: Minimal, because it uses open-source frameworks and existing tools, reducing development and licensing expenses. Scope: On scope	Strength: Flexible development can be combining best features of multiple frameworks. Weakness: More complex to manage and integrate. Opportunities: Can adapt to future system changes easily Threats: Possible compatibility issues between technologies.	Knowledge Gap: Medium requires understanding of multiple frameworks. Skill Gap: Currently limited experience among team members. Technology Gap: Uncertain familiarity with some technologies used in the hybrid approach.	Accepted
Monolithic Architecture	Time: Plausible Cost: Low because it uses single	Strength: Simple to develop, test and deploy Weakness:	Knowledge Gap: None Skill Gap:	Rejected

	codebase and simple to deploy Scope: Partially on scope but not suitable for scaling complex features	Difficult to scale and maintain as system grows Opportunities: Suitable for small-scale applications Threats: Becomes inefficient when integrating multiple components like AI, IoT, and real-time systems	None of the team ever worked with Monolithic Technology Gap: Low as tools are widely available	
Microservices Architecture	Time: Plausible Cost: High because of containerization and monitoring overhead Scope: On scope	Strength: Scalability, modularity and fault isolation Weakness: Requires advanced DevOps, CI/CD pipelines and orchestration tools Opportunities: Could support future expansion if the platform grows Threats: Higher infrastructure costs and security risks from multiple communication channels between services	Knowledge Gap: Moderate as the team had cloud experiences but requires further learning Skill Gap: Moderate as the team had cloud experiences but only for specific objectives Technology Gap: Moderate as cloud services usually requires cost	Rejected

Table 5. Solutions analysis for Backend/Software Framework Component

AI model

Potential Solution	Project Forces	SWOT	KoST	Final Decision
CNNs	Time: Plausible Cost: Medium Scope: Out of scope because it is limited accuracy for abnormal activity detection	Strength: Proven accuracy and efficiency Weakness: Requires a large amount of video data Opportunities: Edge AI deployment is feasible Threats: Dataset quality affects accuracy	Knowledge Gap: Low as the team has AI model training experience Skill Gap: Moderate as there are already pre-trained models but still require some training Technology Gap: High as computation powers heavily affects the training process and efficiency	Rejected
Autoencoder	Time: Plausible Cost: Medium Scope: Out of scope because it doesn't work for activity recognition	Strength: Good for unsupervised and does not require labelled data. Weakness: Difficult to capture complex human action Opportunities: Suitable for small group of datasets Threats: Detection may be unreliable in real-time	Knowledge Gap: None of the team ever worked with Autoencoder Skill Gap: Moderate as we have experience with unsupervised learning but do not ever work with Autoencoder Technology Gap: Low as tools are available but less commonly used in real-time systems	Rejected

Vision Transformer (ViT)	Time: Plausible Cost: Medium-High Scope: On scope because it works well for real-time activity detection	Strength: High accuracy with large datasets Weakness: High computation power required Opportunities: Shown state-of-the-art results when trained with large dataset Threats: Higher difficulty to implement into edge devices	Knowledge Gap: Low as the team had experience on training ViT models Skill Gap: Moderate as there are already pre-trained models but still require some training Technology Gap: High as computation powers heavily affects the training process and efficiency	Accepted
Traditional Machine Learning	Time: Plausible Cost: Low because it can be work on CPU Scope: Out of scope because it is not suitable for complex abnormal activity recognition	Strength: Easy to implement and low computational requirements Weakness: Cannot automatically learn three-dimensional patterns Opportunities: Could be used as fast prototype Threats: Cannot scale to complex movements or subtle abnormal behaviours	Knowledge Gap: Low due to basic ML concepts are sufficient Skill Gap: Low because feature engineering required but manageable Technology Gap: Low because only required standard hardware and no ML specific infrastructure needed	Rejected

Table 6. Solutions analysis for AI components

5.1. Software Diagram

Edge device is identified as Body-worn camera which in our proposed solution is simulated using a webcam. It is assumed to be worked under network connectivity. In the body-worn camera, an AI model is trained and implemented to detect abnormal activities during the guided tour and hence trigger alert and notification is sent to the park guide through the body-worn camera. The data fetched from the AI processing part is stored into the server, including the video streamed by the body-worn camera and the alert notification to be known by the admin through the system. For the front-end part, mainly includes user dashboard and admin dashboard. Information including user profiles, training modules, assessments, certification and alerts are assumed to be stored in the database.

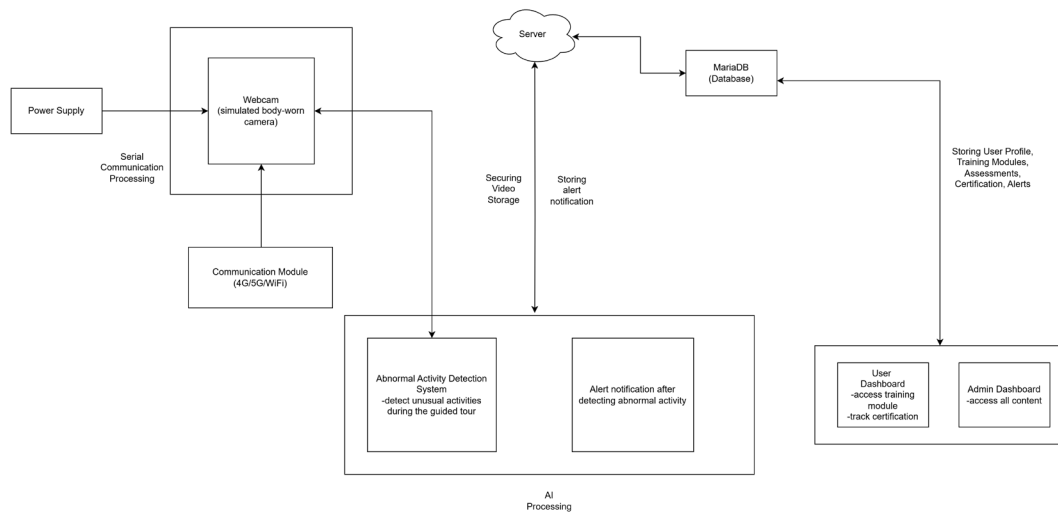


Figure 2. Software Diagram

5.2. Use Case Diagram

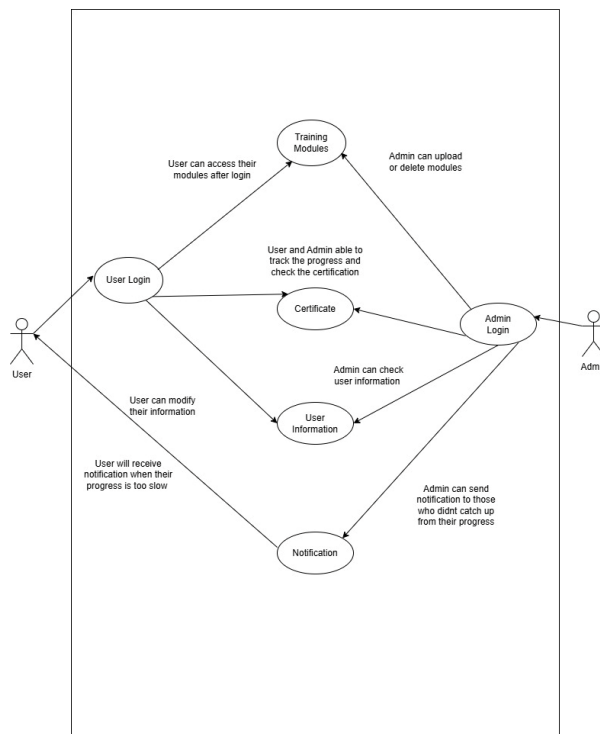


Figure 3. Use Case Diagram

5.3. Activity Diagram

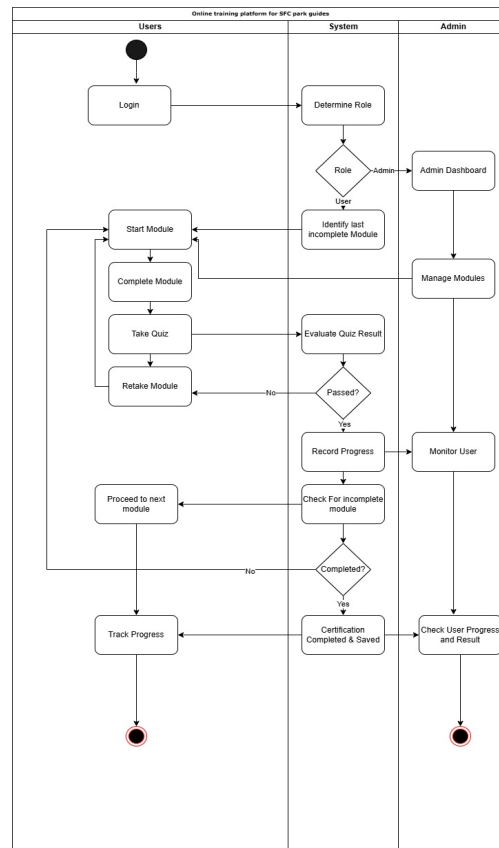


Figure 4. Activity Diagram

6.0. Quality Management

Quality in the context of this Digital Park Guide Training Platform project means delivering a secure, user-friendly, and reliable system built with React Native (frontend), Node.js and Express.js (backend), and MySQL (database). The platform must enable park guides to access up-to-date training modules flexibly through both mobile and web platforms, while the integrated AI-based monitoring ensures real-time detection of abnormal activities during guided tours to support conservation enforcement by Sarawak Forestry Corporation. The system must protect sensitive user data, provide accurate training content, and function effectively even for users with limited technical experience, without compromising performance or security.

Quality will be measured using the SMART criteria defined below, with clear targets applied to every backlog item. These criteria ensure that each feature is evaluated objectively and can be verified at the end of its sprint. In addition, the team will follow the mandatory Secure Software Development Lifecycle (SSDLC) process. After core features are completed in Sprint 1, a full vulnerability assessment will be conducted on the Node.js and Express.js backend APIs, React Native frontend communication, and MySQL database connections using automated security scanning tools. All high or critical issues will be fixed immediately, followed by a reassessment with before-and-after reports as proof. This meets the cybersecurity and data protection requirements from Sarawak Forestry Corporation.

The table below details the Quality Attributes, Definition of Done, and Justification for each backlog item. All measurements are SMART (Specific, Measurable, Assignable, Realistic, and Time-bound) and will be tested by the assigned team member before the end of the respective sprint (Sprint 1 for items 1–5, Sprint 2 for items 6–8). Only items that fully meet the Definition of Done will be accepted as complete.

6.1. Quality management for each backlog items

In a large project, it is important to have a clear understanding of each task to be done and define the definition of done for each quality attributes for the tasks in order to deliver a high-quality product. By doing this, all team members are able to align their goals and avoid possible confusion for the vision of the project. Our sprint goal for each sprint is to achieve every definition of done for each quality attributes for the backlog items in time in order to reserve time for refining the product. The table below shows the quality attributes for each product backlog items and definition of done of them:

No.	Product Backlog Item	Quality Attribute	Definition of Done	Justification
1	User and admin dashboard	Operability	There is a functional and user-friendly interface for both normal users (park guides) and system admin where a new user needs to be able to access their training modules within 5 minutes	User needs to be able to access their training modules and check their user information including certificates and qualifications while admin needs to be able to add and modify training modules and user information without much prior knowledge of similar systems.
2	Training modules	Accessibility	Park guides must be able to comprehend the contents of the training modules without major support within the suggested time shown in the module	Park guides without prior knowledge of similar systems need to be able to understand and complete the learning modules by themselves to learn effectively.
		Appropriateness	Training materials must be accurate and provides credible sources for the information used	Park guides must receive accurate knowledge to avoid any confusion then they are doing their job including providing information for visitors of the park.
3	User and Admin account system	Reliability	3 different accounts are created and tested with different conditions including correct and	The system must have reliable and secure logins for all users to avoid inappropriate access of user data, thus

			incorrect passwords for logins	invalid attempts to login must not allow individuals to access the accounts.
		Capacity	50 mock accounts are successfully inserted into the system database without error	The system must be able to handle a moderate number of users.
4	User data secure and encryption	Security	Sensitive user data are encrypted and unable to access without logging into their own account	Ensures protection of private information and prevents unauthorised access for data including personal info, login credentials and training progress.
		Confidentiality	Access control is enforced from end-to-end and only authorised individuals can access the related data	Confidentiality is key for both admin and user operations, preventing leakage of personal information or training results.
5	AI model training	Accuracy	AI model achieves at least 60% accuracy using test dataset	Ensures the AI performs reliably since incorrect detections could cause situations like false alarm, wasting time of park guides.
		Performance	Model processes and identifies anomalies with latency not exceeding 5 seconds in real-time testing	Low latency for detection can save more time for park guides during guided tour since it allows the system to react as soon as possible.
6	AI-based anomaly detection system	Functionality	AI provides feedback to user within 2 seconds after detected anomaly in real-time testing	Low latency in this context allows park guides to take action to halt any inappropriate behaviours of park visitors.
7	User notification system	Timeliness	System successfully delivers notification within 5 seconds after being triggered	The system needs to be responsive to ensure users always stay updated to their learning and have a clearer grasp of the condition of the park.
		Accessibility	User can access the notification received within 30 seconds without major issue	Ensures the notification is easily readable and accessible to avoid

				confusing park guides and admin.
8	Automated alert system	Responsiveness	Detected anomaly will be reported to park guide and admins within 20 seconds	The system needs to be responsive enough to allow park guides and admin to act as soon as possible to handle the anomaly occurred.
		Availability	Records 5 anomaly in the span of 10 minutes and send alert signal successfully	The system needs to record anomaly and report to park guides and admin and also save the timestamp of the recorded anomaly to allow subsequent review of the footages.
9	Database backup/restore	Reliability	System successfully backup and fully restore all original data during testing	The system must be fully backed up in case of system failures to ensure crucial information such as training progress are not lost.

Table 7. Quality Attributes for each backlog items

7.0. Resources

The hardware and software resources listed below support the full implementation of all mandatory areas of focus for this project. These resources enable the development of the Interactive Digital Training Platform, Cybersecurity and Data Protection measures in compliance with the Secure Software Development Lifecycle and the IoT-Based Abnormal Activities Detection system using body-worm cameras. They align with the hybrid application architecture, Edge-AI processing approach and Scrum agile SDLC model adopted for this project.

7.1. Software Resources

- React Native integrated with the Expo framework is used to develop the cross-platform mobile application and responsive web-based interface. This provides a consistent user experience for park guides and administrators, supporting modular training content, multimedia resources, progress tracking, certification management, and notifications.
- Node.js with Express.js is used to implement the server-side logic, Representational State Transfer (REST) APIs, secure authentication, training module management, and integration with the AI detection system.
- MySQL is used as the relational database to store user profiles, training records, assessment data, certifications, and encrypted video evidence. It supports the required encryption and access control features.
- Jira is used to manage product backlogs, sprint planning, Gantt charts, and the two-sprint release schedule. This supports the Scrum agile methodology and the structured development steps.

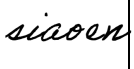
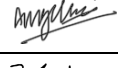
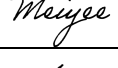
- Visual Studio Code as the integrated development environment and Postman for API testing are used to ensure code quality and adherence to the defined quality criteria.
- The Android Studio emulator and Expo Go app are used as testing tools for the React Native mobile application.

7.2. Hardware Resources

- High-performance laptops and desktop computers provided by team members are used for all coding, database configuration, AI model training and testing, and development activities.
- Webcams are used to prototype real-time video streaming, Edge-AI abnormal activity detection, local alert generation through vibration feedback, and secure video recording and storage. This hardware addresses limited internet connectivity in remote park environments.
- Android and iOS mobile phones are used to install the Expo Go app and test the React Native application across platforms. This ensures reliable validation of flexible access to training modules and notifications for park guides

Approval Signatures:

Project Team

	Name of student	Student Id	Signature	Roles	Description
1	Yi Fei Xavier CHUA	104391384		Project Manager, Developer	Team leader, focuses on backend API and server-side development and cybersecurity measures used in the project
2	Jia Long YEO	104392688		Developer	Focuses on cybersecurity system used in the project and integration of AI system into software and IOT devices
3	Yu Xuan WONG	104392549		Developer	Focuses on AI integration into software and IOT devices and assists on frontend UI development
4	Siao En YONG	104399210		Scrum Master, Developer	Focuses on backend API development and provides assistance on frontend UI development
5	Angeline Hui Lii CHIU	104383462		Developer	Focuses on frontend UI development and use of API in frontend
6	Mei Yee LAI	104390268		Developer	Focuses on database system and AI model training
7	Chuan Kit CHAN	104389635		Developer	Focuses on database system and AI model training

Project Sponsor [Your Tutor]

Tutor's name (on behalf of the client)	Signature:
Ms. Sue Han LEE	

8.0. References

Landsmann, V 2025, 'What is a Project Deliverable? Examples + Workflows', monday.com Blog, viewed 16 March, 2026, <<https://monday.com/blog/project-management/project-deliverables/>>.